

Peer Review

Review of: "Beyond Perplexity: The Quality–Memory Frontier of Native Mixed-Precision KV Caching in llama.cpp"

Evgenii Arsentev¹

1. Independent researcher

1. What the work claims

The paper adds native packed mixed-precision KV-cache types to a llama.cpp fork (Q3 in 32- and 64-element blocks, and a 2-bit value format with a fixed four-level non-uniform codebook, NU-Q2), wires them through CPU, cache-write and CUDA FlashAttention paths, and then asks what the quality–memory trade-off looks like on Qwen3-4B-Instruct (Q4_K_M weights) on one 8 GB RTX 3070 Laptop GPU. Two policies are proposed. The conservative one, Q4K-Q3V, stores 35,712 B/token, a 13.89% saving over Q4K-Q4V (Table 2), and its paired differences on GSM8K, HumanEval and MBPP all have intervals spanning zero (Table 5). The aggressive one, Q4K-MixV-21 (C2), stores 32,768 B/token, a 20.99% saving, slightly lowers aggregate perplexity (Table 4, Wiki PPL 12.2435 vs 12.2919), but loses 2.81 points of strict GSM8K accuracy with paired CI [−5.53, −0.08] and exact McNemar $p = 0.047$ (§5.2). A key-precision ablation (Table 3) shows Q3-64 keys raise PPL by about 20% and Q2 keys collapse, while a 200-example GSM8K pilot favours Q8 keys over Q4 keys by 13.0 points (77.5% vs 64.5%, $p = 0.0007$). The general thesis is in the title and the last sentence of §10: perplexity-preserving KV compression is not necessarily behaviour-preserving, so cache precision should be chosen on downstream and long-context evidence.

I want to credit several things up front, because they are uncommon. The byte accounting is exact: I recomputed every row of Table 2 from §3.2 ($L = 36$, $H_{kv} = 8$, $d = 128$) and Table 1, and all six B/token values and savings reconcile; for C2, keys $36 \times 1,024 \times 0.5625 = 20,736$, values $4 \times 1,024 \times 0.5625 + 4 \times 1,024 \times 0.40625 + 28 \times 1,024 \times 0.28125 = 2,304 + 1,664 + 8,064 = 12,032$, total 32,768. The VRAM measurements in §5.4 match the model too: $8,704 \text{ B/token} \times 32,768 \text{ tokens} = 272.0 \text{ MiB}$, exactly the

reported 32K gap, and $4,188 - 3,916 = 272$. The downstream percentages are consistent with integer counts (Table 4, GSM8K column: $65.43\% = 863/1,319$, $62.62\% = 826/1,319$, difference $-37/1,319 = -2.81$). And Table 6, the failure ledger, is the most honest section I have read in a KV-quantization paper in some time. A phantom uniform-Q2 control, "long-context" prompts that were only hundreds of tokens long, a C2 map that silently ran as uniform Q4: these are exactly the bugs that quietly produce published results elsewhere. Declining to copy numbers from other papers into a fake head-to-head (§2) is also the right call.

2. Whether the conclusion holds on the data

The headline behavioural claim for C2 rests on one borderline test outside the declared confirmatory family. §4.4 declares the confirmatory family as Q4K-Q3V on GSM8K, HumanEval, MBPP and 15 RULER cells, with Holm correction, and says that "exploratory all-policy comparisons are not promoted to confirmatory claims." The C2 GSM8K result is not in that family. Yet §5.2 calls it "a borderline pre-specified result", and the abstract and §10 present it as the central demonstration. I could not find where C2 versus Q4K-Q4V was pre-specified. If C2 had its own family of the same 18 tests, the smallest Holm threshold would be $0.05/18 = 0.0028$; even restricted to the three downstream tasks it would be $0.05/3 = 0.0167$. $p = 0.047$ clears neither. The paper half-concedes this ("would not survive an indiscriminate correction"), but the abstract does not.

The thesis itself is not lost, in my reading. The stronger evidence for "similar likelihood, different behaviour" is actually on the key axis: Table 3 gives Q8 and Q4 keys nearly identical PPL (Wiki 12.3683 vs 12.3227) with a 13.0-point GSM8K gap at $p = 0.0007$, and Table 4 shows the same direction at full size, Q8K-Q8V 75.21 vs Q4K-Q4V 65.43 on GSM8K (992 vs 863 of 1,319, a 9.78-point difference) while Wiki PPL differs by 12.2680 vs 12.2919. I would rebuild the argument around that contrast and report C2 as supporting, exploratory evidence.

The frontier is conditional on Q4 keys, and the paper's own data say keys matter more than values. Every proposed policy fixes keys at Q4 (§3.2). But the largest downstream effect in the paper is the Q8-vs-Q4 key gap, roughly three to five times the C2 value loss (9.78 or 13.0 points against 2.81). Moving keys from Q4 to Q8 costs $36 \times 1,024 \times (1.0625 - 0.5625) = 18,432$ B/token by the paper's own accounting; C2 saves 8,704. A reader deciding where to spend memory needs Q8K-Q4V (59,904 B/token) and Q8K-Q3V (54,144 B/token) evaluated on full GSM8K, HumanEval, and MBPP. Without them, calling Q4K-Q3V and C2 a "two-point deployment frontier" (abstract) overstates what was mapped: it is a frontier along the value

axis at a fixed, and apparently costly, key precision. There is also no FP16 cache reference anywhere. Under Eq. 1 with 2 bytes per FP16 element, the default cache would be $36 \times 8 \times 128 \times 4 = 147,456$ B/token, so Q4K-Q4V is already a 71.9% reduction whose quality cost is unmeasured.

"Statistically indistinguishable" is not the same as equivalent. Table 5 reports Q4K-Q3V minus Q4K-Q4V intervals of [-1.06, 4.32] on GSM8K, [-4.88, 8.54] on HumanEval, and [-2.80, 4.40] on MBPP. The GSM8K interval rules out a loss larger than about one point, which is good. HumanEval does not rule out a loss of nearly five points, on 164 problems. §5.2 wording ("quality-preserving in the tested samples, not quality-improving") is careful; the abstract's "statistically indistinguishable" and Table 7's "Reason/code sensitive" recommendation lean further. A pre-stated non-inferiority margin, or simply putting the lower bounds into the abstract, would fix this.

Aggregate PPL differences have no uncertainty attached. The abstract says C2 "slightly lowers held-out perplexity." From Table 4, that is $12.2435/12.2919 - 1 = -0.39\%$ (Wiki) and $18.3585/18.4667 - 1 = -0.59\%$ (OWT). §4.2 says a paired OpenWebText chunk analysis exists "only for its paired confidence interval," but I could not find that interval reported. The same table has Q8K-Q8V with a worse OWT PPL than Q4K-Q4V (18.5222 vs 18.4667), which suggests differences of this size sit inside run-to-run or chunk-level noise. Without the interval, "slightly lowers" and "slightly improves" (§10) should become "does not measurably change."

Three values look like possible copy or pipeline errors, and each touches a claim.

- Table 4, HumanEval column: All-NU-Q2 V and Uniform-Q2 V both show 65.85 (108/164), while the same two rows differ by 27.52 points on GSM8K (58.53 vs 31.01) and 17.20 on MBPP (48.00 vs 30.80). Identical code scores for a collapsed and a non-collapsed cache are possible, but given that Table 6 records a "Uniform-Q2 phantom" where the control reused the non-uniform kernels, I would want the per-problem outcomes and output fingerprints for these two HumanEval runs specifically.
- §6.4: on Qwen3-1.7B, "GSM8K falls from 65.43% to 62.02%." 65.43% is exactly the Qwen3-4B Q4K-Q4V GSM8K score in Table 4. Two different models landing on the same 863/1,319 is not impossible. It is unlikely enough that I suspect a transcription error.
- Appendix B (Table 9), Q4K-Q4V variable tracking: 52 at 8K, 7 at 16K, 99 at 32K. A baseline that scores 7% at 16K and 99% at 32K on the same task points to something about the 16K generator or its prompts rather than model capacity. The C2 "+60 points at 16K" in §5.3 is measured against this cell. Q4K-Q3V (61, 14, 100) shows the same dip. Multi-key rising with context for the baseline (32.75, 35.25, 49.25) is odd too.

The paper's discussion of RULER (§5.3) already refuses to read the positive cells as "compression increasing intelligence", which I appreciate. But "task- and context-dependent changes rather than monotonic degradation" in the abstract is a claim about quantization, and it is hard to separate from an artifact of the evaluation code until the 16K baseline dip is explained. There is also no high-precision (Q8K-Q8V or FP16) RULER row to show what the model does at 16K with an uncompressed cache. The "empirical self-resampling noise floor" mentioned in §5.3 is never given as a number.

GSM8K scoring may conflate formatting with reasoning. §4.3 uses "strict first-attempt exact match." Quantization changes token trajectories (§5.3 says so), and a strict extractor penalises a correct answer printed in a different format exactly as it penalises a wrong one. The paper interprets the C2 loss as reduced "mathematical reasoning" (§1, contribution 3). Reporting a flexible-extraction score alongside, or the share of failures that are unparseable, would show which of the two it is.

The layer-selection heuristic is never described. §3.2 gives the C2 map {0, 7, 20, 23} at Q4, {9, 14, 16, 34} at Q3-64) and §6.3 calls it "the heuristic", but I could not find what the heuristic is: a sensitivity score, a PPL search, something else. §6.3 reports that C2 beats 30 of 32 random equal-memory maps on selection PPL, and that Random-23 did better. Given that the paper's main message is that PPL does not predict behaviour, selecting (or validating) a map on PPL deserves a sentence of reflection. Random-23 appears in RULER (Table 9) but its GSM8K score is not reported, which is the one number that would test whether the choice of map matters for the mismatch.

The systems claims hold. Memory saving matches the analytic model to the MiB, and the 6.5% process-level reduction (272/4,188) is correctly distinguished from the 20.99% KV-only figure. One clarification is needed: in llama.cpp, the cache is normally allocated for n_ctx up front, so a measured gap that equals the analytic value exactly at every length suggests n_ctx was set equal to the populated length. That should be said, since "populated context" otherwise implies something different. Throughput changes of 1–3% are reported only as relative values; absolute tokens per second would let readers compare with their own setups.

3. Reproducibility

Better than typical on hardware and environment: GPU, CPU, CUDA toolkit 13.3, driver 610.43.03, CMake 4.4.0, and Python 3.13 are all recorded (§4.1), the evaluation sizes are stated (§4.3), and the code is linked. The following are missing, and several are needed to reproduce the headline numbers.

Model artifact. "Qwen3-4B-Instruct with Q4_K_M weights" does not identify a checkpoint. Qwen3-4B was released as a hybrid thinking model and later as separate instruct variants; the paper should give the exact Hugging Face repository and revision, the GGUF file source and its hash, and whether thinking mode was on or off. The same applies to Qwen3-1.7B and Llama-3.2-1B-Instruct.

Code version. The Data Availability section gives a fork URL only. No commit hash, no upstream llama.cpp base commit, no branch or tag. Evaluation tooling (the rebuilt RULER generators, the corrected HumanEval/MBPP execution code listed in Table 6, the GSM8K extractor) is not said to be in the repository.

Decoding. "Deterministic pass@1" suggests greedy decoding, but temperature, top-k/top-p, maximum new tokens, stop strings, chat template, and system prompt are not given for any benchmark. The GSM8K prompt (zero-shot or few-shot, and with which exemplars) is not stated.

Data files. §4.2 says calibration, selection, and evaluation files are disjoint, which is the right design, but the files themselves are not identified: which WikiText split and version, how the OpenWebText test file was drawn, and how large the calibration and selection sets were.

Statistics. The number of bootstrap resamples and the resampling seed are not stated. Holm-adjusted p-values are described in §4.4, but Table 5 reports only raw p. The threshold used for the "thresholded pass/fail" RULER analysis (§5.3, $p = 0.0013$) is not defined.

Codebook details. Eq. 2 gives the levels but not the scale convention: what is s ? If s is the block max-abs, the outermost level at 0.6193 means every block maximum is clipped, which would be worth stating. §3.1 and §6.2 refer to "rotated deployment values" and "post-rotation value windows," but no rotation is described anywhere in §3. Values do not receive RoPE, so I assume a different transform is meant; it needs to be specified. The small asymmetry in the levels (-0.619326 vs 0.619301) and thresholds (-0.402845 vs 0.402832) is also unexplained. The thresholds themselves are correct midpoints: $(-0.619326 - 0.186363)/2 = -0.4028445$.

Figures. The text I reviewed on Qeios does not carry the content of Figures 1–6; I could not check the RULER deltas plot (Figure 4), the systems plot (Figure 5), or the calibration plot (Figure 6) against the tables, and my comments rely on the tables and prose only.

Raw outputs. Per-example generations and scores are not released. For a paper whose own ledger (Table 6) shows how easily a control can silently run the wrong kernel, per-example outputs with binary hashes are the most useful artifact the author could add.

4. What specifically to fix

1. **Abstract, §5.2 and §10.** Stop calling the C2 GSM8K result "pre-specified" unless a pre-registration or earlier plan can be cited; §4.4 places it in the exploratory set. Report it as exploratory, $p = 0.047$, and lead the likelihood–behaviour argument with the Q8-vs-Q4 key evidence (Table 3, GSM8K-200, 77.5% vs 64.5%, $p = 0.0007$; Table 4, GSM8K, 75.21 vs 65.43).
2. **Table 4 and Table 7:** add Q8K-Q4V and Q8K-Q3V on full GSM8K, HumanEval, and MBPP, plus an FP16-cache row. Without these, the "frontier" is only the value axis at Q4 keys.
3. Table 4, HumanEval, All-NU-Q2 V and Uniform-Q2 V (both 65.85): confirm from per-problem outcomes and binary hashes that these are two distinct runs.
4. §6.4, Qwen3-1.7B GSM8K baseline of 65.43%. Please check it; it is identical to the Qwen3-4B Q4K-Q4V value in Table 4.
5. **Appendix B, Table 9, variable tracking at 16K** (Q4K-Q4V 7, Q4K-Q3V 14). Explain the dip, or regenerate the 16K cell and inspect prompts for a generator difference. Add a Q8K-Q8V RULER row. Until then, remove the "+60 points at 16K" from §5.3 or mark it as unexplained.
6. §5.3: give the numeric value of the "empirical self-resampling noise floor" and say how it was estimated.
7. **Table 5.** Add a Holm-adjusted p column, as §4.4 promises, and state the family size. For the conservative policy, consider a non-inferiority framing with a margin chosen in advance; the HumanEval interval $[-4.88, 8.54]$ cannot currently exclude a nearly five-point loss.
8. §4.2 and Table 4 PPL columns: report the paired OWT chunk CI for C2 and Q4K-Q3V. Replace "slightly lowers/improves perplexity" (abstract, §5.2, §10) with the measured change and its interval.
9. §4.3: add a flexible-extraction GSM8K score, or the count of unparseable answers per policy, so the C2 loss can be attributed to reasoning rather than formatting.
10. **§3.2 and §6.3.** Describe the heuristic that produced the C2 map. Report Random-23 on GSM8K, since it beat C2 on selection PPL.
11. Eq. 2 and §6.2: define the block scale s , the rotation applied to values before calibration, and the reason for the slightly asymmetric levels.
12. §4.1 and Data Availability: exact model revisions and GGUF hashes, thinking-mode setting, fork commit hash and upstream base commit, decoding parameters, prompts and templates, bootstrap resample count and seed, and the RULER pass/fail threshold.

13. §5.4: state `n_ctx` for each measured length and add absolute prompt-processing and generation throughput.
14. Cross-references in §5.1 ("Tables 3 and 2 closes") and §5.2 ("Tables 4 and 3 presents") are broken, apparently left over from renumbering. Table 4, TQA BLEU for Q4K-Q3V is "-" with no explanation.
15. Table 3 footnote: 2,151.9 is " $174.6 \times Q4 \text{ PPL}$ " ($2,151.9/12.3227 = 174.63$), but the Q2 value uses 16 chunks and the Q4 value 64. Compute the ratio on the same 16 chunks or say that the subsets differ.

5. Recommendation

Major revisions. The engineering contribution is real and carefully accounted for: native packed formats in a widely used runtime, byte and VRAM numbers that reconcile exactly, paired statistics, and an unusually candid failure ledger. The central claim that perplexity is an unreliable selector for KV precision is probably right, and the key-precision data already support it better than the C2 result does. What keeps me from minor revisions is that the headline C2 finding is exploratory and borderline while being presented as pre-specified, the frontier omits the key-precision rows that the paper's own data say matter most, three table values (identical HumanEval scores for the two Q2 controls, a repeated 65.43% across models, and a 7% baseline at 16K variable tracking) need to be checked before the corresponding claims can stand, and the model revision, commit, and decoding settings needed to rerun the work are not given.

Declarations

Potential competing interests: No potential competing interests to declare.